

Overview

Using a Raspberry Pi as its Brain and 6 stepper motors to control the arms, the Rubik's Cube Solving Robot is able to visually capture, analyze and solve a 3x3 Rubik's Cube.

Software

I developed and utilized many programs to analyze and solve the Rubik's Cube. The main tasks that had to be accomplished with my code included capturing the current state of the cube, analyzing the current state to generate a solution, and the execution of the solution onto the physical cube.

```
print("UP side colors.")
p0=(0,0),(0,0),(0,0),(0,0),(0,0),(0,0),(0,0),(0,0)
p01=(176,395) #U2
p02=(271,317) #U2
p03=(392,186) #U3
p04=(263,771) #U4
#US WHITE BY DEFAULT
p05=(549,315) #U6
p06=(340,973) #U7
p07=(480,887) #U8
p07=(634,634) #U9
```

Background

I've always been fascinated with Erno Rubik's 3x3 puzzle cube. I learned how to solve them when I was 12 and even participated in competitions. I am also a computer science student with an interest in robotics and automation. This project is a culmination of my passions and countless hours of diligent work.

Color Detection

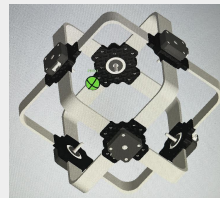
The cameras on the robot are able to take a picture of the cube. Using a color detection algorithm on each cubie it is able to capture the current state of the Rubik's Cube. This information is then passed to the solving algorithm.



CAD/3D PRINTING

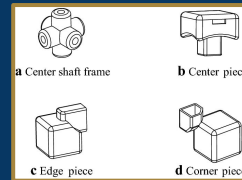
Using the Computer Aided Design program Inventor, I designed the frame of the robot along with the robotic arms that manipulate the cube.

Using a 3D printing slicer along with a 3D printer I was able to manufacture all of the parts that I had designed.



Rubik's Cube

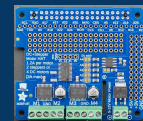
The Rubik's Cube is a fascinating puzzle that often perplexes those unfamiliar with its internal mechanics. In understanding the mechanical structure the cube it becomes easier to manipulate.



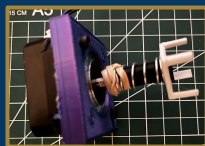
Raspberry Pi

The Raspberry Pi is a low cost computer that is highly adaptable to a wide variety of applications. In this project the Raspberry Pi is configured with 2 usb cameras and 3 motor-controller boards.

This custom configuration is what drives the robot. It handles the image capture, computations and mechanical action.



Working Photos



Python/ Python Libraries

All of the code in this project is written in python. Python is a very powerful programming language with lots of libraries. Those libraries make it much faster for me to write the code necessary for this specific project.